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THE ROLE OF GANTT CHART IN PROJECT MANAGEMENT

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ABSTRACT

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The main objective and goal of this study is to know how the Gantt chart can assist project managers to communicate and collaborate in an effective way. Literature highlights that most powerful tools available for this purpose is the Gantt Chart.

This thesis explores the utilization of Gantt charts in project management, emphasizing their role in planning, scheduling, and monitoring project progress. The research method used in the study are literature review in the theoretical section, and case study methodology in the empirical section.

The results show that the Gantt chart can facilitate the visualization of project timelines, task dependencies, and resource allocation in enhancing project coordination and communication among different industries. Additionally, the study addresses common challenges and limitations associated with the Gantt chart implementation, such as inaccurate time estimates, resource allocation issues, and lack of flexibility and offers best practices for overcoming these obstacles. The findings underscore the importance of adopting Gantt charts in modern project management to ensure timely and successful project delivery.

Keywords Gantt chart, project management, project communication, Gantt chart tools, project collaboration.

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1 INTRODUCTION

The GANTT (Generalized Activity Normalization Time Table) chart is a bar chart developed as a project management tool that was proposed by Henry Gantt. Frederick W. Taylor discovered the limitations of Gantt chart traditional methods where they truly explain the method of planning and execution of work Darmody et al. (2007). "Gantt chart" is a conceptual root of project management tool Geraldi et al. (2012). The Gantt chart is one of the most important tools involves planning, organizing, controlling resources, achieving specific goals within constrain of time, budget and scope of the projects. According to the researched survey with 750 project managers, the Gantt chart was the fourth most used tools out of 70 tools and techniques associated with project management (Besner and Hobbs, 2008). (Millerson and Wilensky, 1964) noted the rapidly increasing number of new occupations seeking recognition as professions. Its simplicity and effectiveness makes it a valuable asset for coordinating and tracking progress in a wide range of fields, including construction projects (Chbaly and Brunet, 2023), software (Dayani and Gelbard, 2015), schedule event (Soare et al., 2019), and manufacturing industries Saravanan et al. (2018).

Without effective project management tool, the most promising projects may face significant challenges or fail to achieve their objectives. It's important to provide productive communication tools to every staff member within the team. To manage the projects, where staff members have multiple responsibilities must have efficient skills to fulfil the projects. According to Lyons and Gische (2005) emphasizes on the Project Management Tools and Techniques. The successful project always relies on the team development that is actively engaged. This concept of project management refers to "soft" or human side to complete project. This thesis discusses the significance of communication tools, team collaboration and interpersonal skills to fulfil the project goals.

To check the performance of the project there are three main key factors scope, schedule and budget. Alvarado et al. (2004) evaluate the success of construction projects; failure of projects are often seen due to poor management mostly at front end during strategy formulation rather than execution. Anderson and Merna (2003) talk about how business strategy should be translated into projects. Archer and Ghasemzadeh's (1999) study highlights being good at the project portfolio means doing all the tasks and duties really well. Many industries like research and development of new products, manufacturing, construction projects follow the project management approaches. The motivation is to not only having tools for their projects but also, it's really valuable for each stakeholder to know the significance of the Gantt chart as a communication tool.

1.1 Research Objectives

The primary aim is to research effectiveness of the Gantt chart within the realm of project management as a communication tool. This study seeks the effectiveness in communication among project stakeholders. To achieve this, our research showcases the best two research objectives. Firstly, it evaluates the importance of Gantt charts as communication tools in project management. This objective involves analysing the clarity, accessibility, and understanding how to convey information through Gantt charts. Secondly, the aim is to find ways to integrate the Gantt chart into the project management workflows and how it helps in communication process. The objective is to examine how the Gantt chart tool is used for creating, sharing and updating tasks within project teams.

1.2 Research Gaps and Research Questions

The Gantt chart is used as a project management tool, nevertheless, there is a gap in research how it influences project performance and communication. To enhance the project outcomes, numerous organizations have explored many project management tools and techniques but specific role of the Gantt chart remains unexamined. Gaps can be observed where the Gantt chart is not very user-friendly

such as integration with AI and automation, real-time collaboration, visualization enhancements, integration with other tools, mobile optimization, integration with virtual reality (VR) and augmented reality (AR). This research will address valuable insights into the effectiveness of project management tool as Gantt charts. Technology has improved, Gantt charts have gotten even better. Now, project managers can handle big and complicated projects with ease using advanced Gantt chart tools. Therefore, the objective of the study is to address the research questions mentioned below:

How can Gantt charts be integrated in project collaboration and communication tools?

1.3 Thesis Structure

The structure of the thesis is presented in Figure 1 providing a clear overview of the research organization. It demonstrates the sequential flow of the study, from introduction, literature review to conclusions. Every section is interlinked to reflect the logical progression of the research.

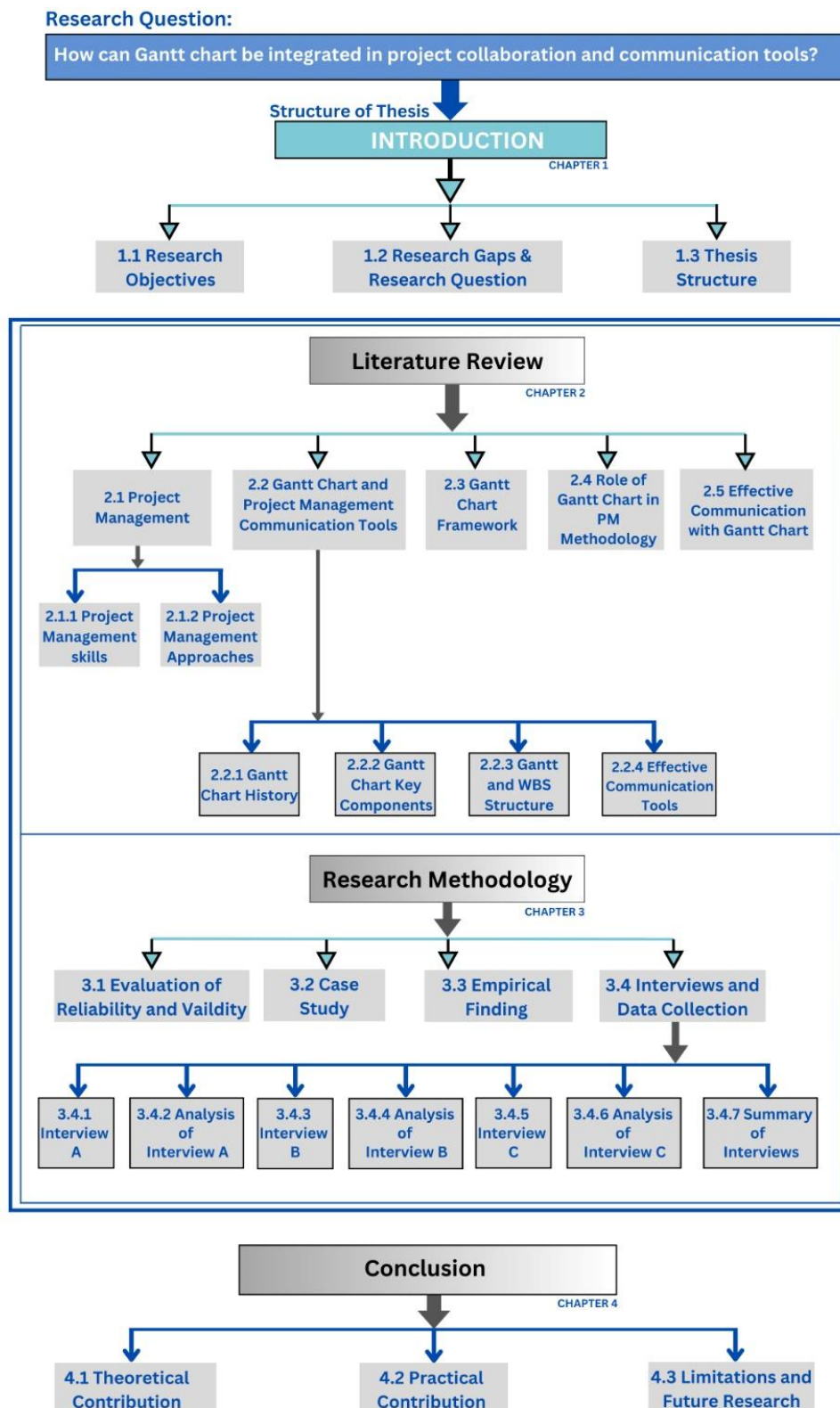


Figure 1. Overview of the thesis structure

2 LITERATURE REVIEW

During the research the keywords used were “Project Management”, “Gantt Chart”, and “Communication & Collaboration” to find best publications and to filter out relevant material for my thesis. Terms method and methodology both are related but distinct concept in research according to (Kirsch and Sullivan, 1992). A term “Method” means technique use to collect evidence, whereas “methodology” means encompasses the underline theory and analysis of the research process. In this chapter, literature review method employed to collect secondary sources which includes the published articles. Our main goal is to gather valid evidence to address our research problem. For this I used Google Scholar, ProQuest, Emerald, ScienceDirect, and Segar Journals for my research.

2.1 Project Management

2.1.1 Project Management Skills

Definition elaborated by the “Project Management Institute” by Seymour (2014) define a project as “a group activity where team members aimed to produce product, services and other outcomes” and Defining “Project Management” as “to fulfil the project requirements it use application of knowledge, skills, tools and techniques to project activities”. For project management skills, there are crucial key elements that encompass effective planning, project execution and completing projects with multiple complexities. Project management is an art. Cash et al. (1992) say that there is no existing formula which showcase that the project will deliver expected outcomes within timelines and budget constraints. Skills play an important role for project managers to navigate the key challenges for successful outcomes. The best key elements for project management skills presented in Figure 2 are leadership, time management, communication, risk management, adaptability, quick problem-solving and good organizational abilities.

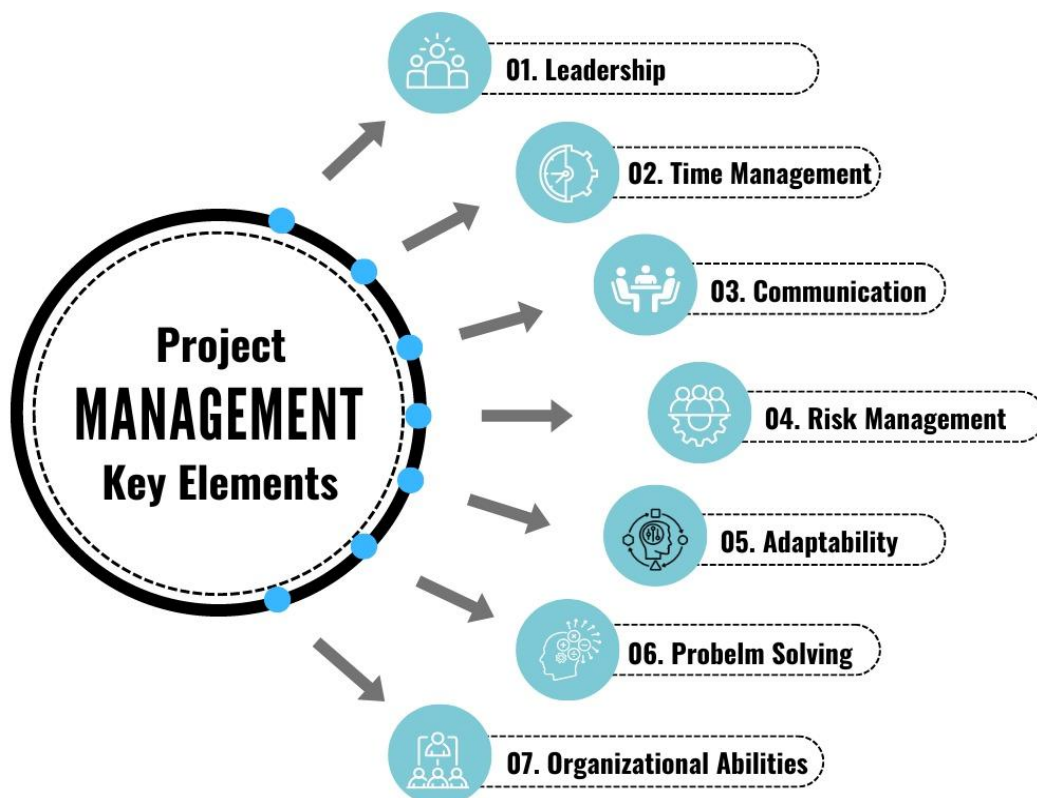


Figure 2. Project management key elements

Leadership the first key element; strong leadership qualities in project managers can easily motivate other team members and help to achieve common team goals. Leadership key elements can be inspiring team members, providing right direction, a collaborative thought and supportive environment. A good leader will always inspire confidently during the uncertain times or drastic challenges and obstacles.

Another important element is **time management**, (Hines, 2010) where project managers often deal with tight deadlines and schedules in the projects. So, to fulfil the tasks on time and within budget, the project manager will allocate the specific time and resources. Simple or complex, the project always has realistic timelines, prioritizing tasks, and monitoring tools to identify the project status or to know potential delays in the project.

Communication skills (Zulch, 2014) are equally vital for every project manager to transfer information and new ideas. For every team member, stakeholders and

clients, it's important to communicate timely to avoid conflicts, misunderstanding, and delays in projects. Communication ensures that every team member is on the same page in regard to project objectives, requirements and timelines. However, active listening skills helps project managers to understand the needs and problems of stakeholders.

The next critical element explained by (Yang, 2013) is **risk management** where project objectives and outcomes are impacted by uncertain risks. Skilled project managers cop up to find the potential risk and then develop strategies to mitigate those risks. To keep the projects on track, many organizations use proactive approach to mitigate the project risks so that projects remain on track.

Adaptability is the next essential skill used especially in automotive industry Kiefer et al. (2017) and which gives all the project managers a dynamic and ever-changing scope of project environment. Every project may need unexpected changes or shifts in requirements. In such case project managers are proactive to understand the changed requirements and flexible handle and capable to adipate all the changes quickly and effectively keeping all things in mins like project goals and objectives.

A skill like **problem-solving** Ahern et al. (2014) resolves various conflicts among the internal or external team members. With the help of the project manager, the root causes of problems faced by team members can be identified. However, the project manager develops creative solutions and implement the effective action plans.

Effective **organisational** skills Pedraza-Rodríguez et al. (2023) involve coordinating multiple tasks, current resources, and skilled stakeholders to achieve desired outcomes. To ensure that all the project objectives are meeting efficiently, project managers take care to organize and priorities tasks and set the budget. Additionally, a strong project manager with strong organizational skills keeps a track record

including project progress, performance monitoring, milestones, and checks if deliverables are providing valuable insights to organization.

2.1.2 Project Management Approaches

To establish the principle of project management knowledge, Project Management Institute (PMI) originated the Project Management Body of Knowledge (PMBOK). According to Marnada et al. (2022), it aims to execute the project through successful project managers by furnishing updated knowledge base and implementing the structured framework for project management. There are six editions of the PMBOK released by PMI, published in 2017. It showcases the ten knowledge areas: Project Integration Management, Project Scope Management, Project Schedule Management, Project Cost Management, Project Quality Management, Project Resources Management, Project Communication Management, Project Risk Management, Project Procurement Management, and Project Stakeholder Management. The break-down of the concept of project management according to Marcelino and Domingues (2022) consists of five groups: initiating, planning, executing, monitoring & controlling, and closing, as depicted in (Figure 3). PMBOK highlight the integrative role of agile and traditional in the project management.

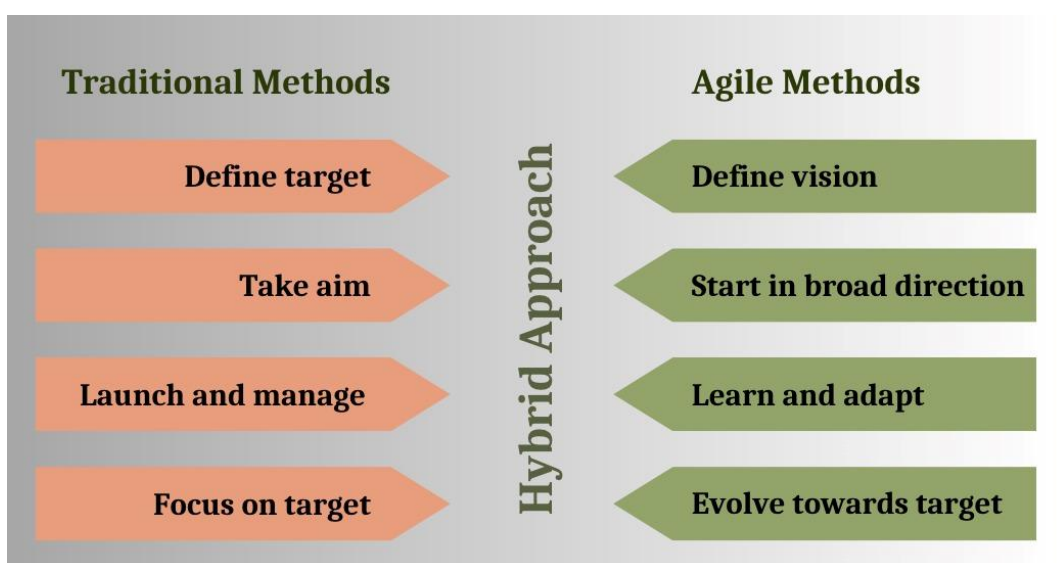


Figure 3. Project management approach.

Multiple growing businesses face hard competition with dynamic opportunities. Before executing the project, traditional project management approaches have a crucial role to plan and anticipate in project details and requirements. According to Ciric et al. (2019), contemporary environment often falls under the swift changes in traditional model. However, organizations must acknowledge the need of adaptability. Extended agile project management (APM) always practices beyond the traditional domain. Agile methodologies focus on flexibility, adaptability to change and collaboration. Špundak (2014) talks about the primary aim of traditional project management approach. To achieve optimization and efficiency is to closely look into the detailed project plan. This approach helps to complete the project within the time constrain, budget and scope.

For example, Fabric-GC gives us a proposed solution that integrates the Blockchain technology with the GANTT chart model for project management Androulaki et al. (2018). The proposed solution, Fabric-GC, addresses these challenges by leveraging the Blockchain technology and the Gantt chart model Leila et al. (2013). Problems faced in managing, controlling and operating can be mitigated with the help of potential blockchain-based solution.

2.2 Gantt Chart and Project Management Communication Tools

2.2.1 Gantt Chart History

Today, Gantt charts serve as essential tools for project managers to communicate schedules, identify potential delays, and focus team efforts on critical tasks for timely project delivery. The historical development of the Gantt chart reveals two key insights. Darmody et al. (2007) *Firstly*, originating within the scientific management movement, the Gantt chart was initially devised to monitor worker performance and resource utilization with a focus on efficiency and rational analysis. Its evolution reflected the ethos of scientific management, emphasizing clear task durations, error correction, and continuous improvement. *Secondly*, while originally

intended for repetitive operations, the Gantt chart found application in project contexts with minimal adaptation, gaining popularity in project management in the 1950s. Despite its roots in scientific management, its universal use across various project types highlights its enduring relevance. Conceptually, the Gantt chart's principles, derived from scientific management, emphasize rational analysis, accountability, and systematic improvement, impacting project management by providing a structured approach to planning and control. The Gantt chart, initially introduced by Gantt in 1903 alongside Frederick W. Taylor's "shop management" paper, has evolved from a production planning tool for batch production to a crucial method for project planning and control. Its widespread use in project management stems from its intuitive visual representation, with Moore and Buffa advocating for its adoption in the late 1960s Winslow et al. (1903). Earned Value Analysis (EVA) is a project management technique originally developed by the U.S. Department of Defence during the 1960s to measure project progress in an objective manner Abba, (1997) Specifically, EVA brings cost and schedule variance analysis together to provide accurate status of a project (Association for Project Management, 2008) . Cost and time forecasts are able to project deviations under different hypotheses Maylor (2001). Although EVA is a PM technique, due to its reliability, it has been widely adopted by several sectors, such as the construction sector, the manufacturing sector, and the information technology sector, Liberatore, Pollack-Johnson, and Smith have highlighted that almost half of the construction industry uses PM tools in construction management, and EVA is one of the most frequently-used tools.

2.2.2 Gantt Chart Key Components

The structure of a Gantt chart is fundamental to understanding and utilizing this project management tool effectively. According to Elizabeth Harrin, (2019) Gantt chart is the visual representation where tasks are interconnected with each other until project is finished. This show that the tasks are interdependent among different tasks. However, every project manager should ensure to complete task on

time, and there should be minimum delays. Here's a breakdown of Gantt chart described by Elizabeth.

Tasks or Activities: These are the individual components of a project, representing work that needs to be completed. Tasks can vary widely depending on the project, but they're essentially the building blocks of the project.

Timelines (Start and End Dates): The timeline of a Gantt chart displays the duration of each task. They provide a clear schedule for when work should begin and when it should be completed.

Task Dependencies: Gantt charts help visualize the relationships between different tasks in a project. For example, one task may need to be completed before another can begin. These dependencies are often represented by connecting lines or arrows between the corresponding bars on the chart, showing the sequence in which tasks must be executed.

Horizontal Axis (Time): The horizontal axis of the Gantt chart represents time. Typically, it's divided into days, weeks, or months, depending on the duration of the project. Each unit of time is marked along the axis, allowing for the visualization of when tasks are scheduled to occur throughout the project timeline.

Vertical Axis (Task Scheduling): The vertical axis of the Gantt chart represents task scheduling. It lists down all the tasks or activities that need to be completed as part of the project. Each task is represented by a separate row or bar along this axis. Tasks are typically arranged in order of priority or sequence.

Bars: The main visual components of a Gantt chart are the bars or blocks that represent the tasks or activities scheduled over time. Each bar typically starts at the scheduled start time of the task and ends at its scheduled completion time. The length of the bars represents the duration of each task.

Milestones: In addition to individual tasks, Gantt charts may include milestones, which are usually represented by distinct symbols or markers on the chart. These can include project start and end dates, major deliverables, or significant project phases. Milestones provide visual markers for tracking progress and measuring project success.

Resources: Some Gantt charts include resource allocation information, indicating which resources (e.g., personnel, equipment) are assigned to each task or activity.

2.2.3 Gantt and WBS Structure

Gantt charts illustrate the start and finish dates for both the smallest tasks in a project and higher-level tasks that encompass multiple terminal elements. The representation aligns with the hierarchical structure of project planning, known as the work breakdown structure (WBS). According to Sharon and Dori (2012), WBS is a hierarchical method used in project management to break down the total scope of work into smaller, more manageable components, enabling effective planning, execution, and monitoring. It starts with high-level tasks, it progressively subdivides the tasks into smaller units, facilitating a clear understanding of project work at each stage. It forms the basis for project management activities such as scheduling, budgeting, resource allocation, and risk management, empowering project managers and team members to effectively manage dependencies, allocate resources, and track progress throughout the project lifecycle (Besner and Hobbs, 2008). To get comprehensive project management framework, to compare WBS with Gantt chart. WBS have critical role in defining the scope of work and organize the project activities into manageable sections. Below is the sample framework for WBS referred in Figure 4.

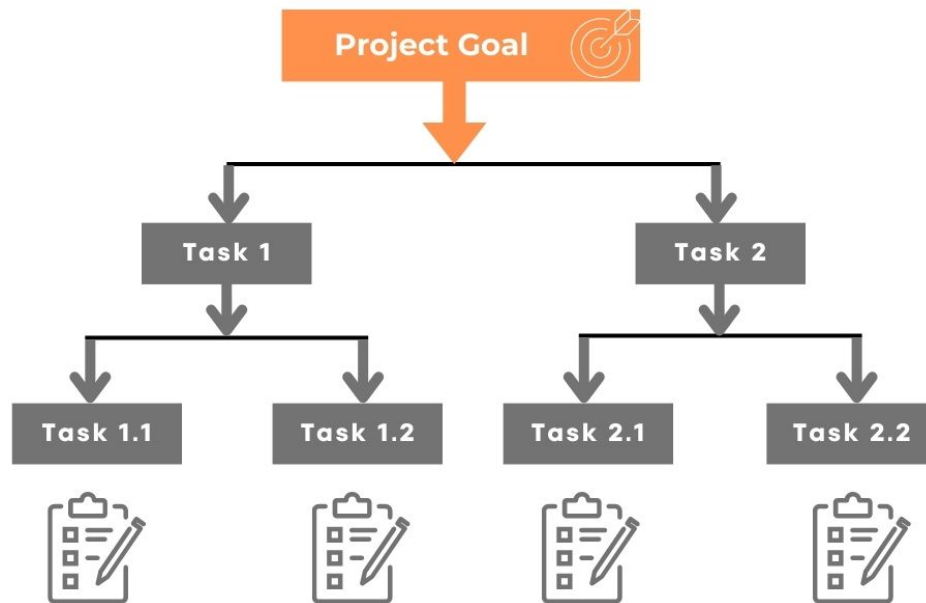


Figure 4. Work breakdown structure example.

Gantt charts can be enhanced to display the current schedule status by shading bars to indicate the percentage of completion. The vertical line representing the current date (see Figure 5), typically labelled as “TODAY” serves as a reference point for comparing the planned schedule to the current date. It helps project managers and team members in assessing whether tasks are progressing as scheduled or if there are any delays. If tasks are behind schedule, adjustments can be made promptly to mitigate risks and keep the project on track. **The chart provides a visual representation of a project’s schedule, with tasks listed on the vertical axis and time intervals displayed along the horizontal axis.**

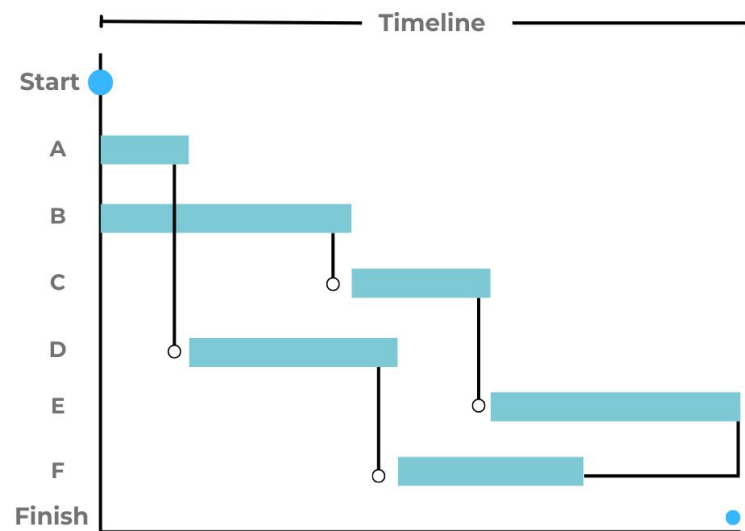


Figure 5. Gantt chart structure example.

However, Gantt chart helps in visualizing the project schedules, timelines and other task dependencies. When these are combined, the project manager initiates their project by planning, organizing, executing and monitoring projects more efficiently.

2.2.4 Effective Communication Tools

There are various specialized management tools and software's that offer advance features to drafting a Gantt chart. With the help of those software interface, that allow users to customize and update their Gantt charts based on their project requirement. To highlight some project management software's Instagantt (2024, May). Top 10 Gantt Chart Software in 2024. Retrieved 2024-07-22, from <https://www.instagantt.com/top-10-gantt-chart-software-for-project-management> mentioned below:

Microsoft Project: Microsoft Project is one of the most widely used project management tools for creating Gantt charts that allows users to create detailed Gantt charts, track tasks, manage resources, and collaborate with team members. It offers advanced scheduling capabilities, resource allocation, and reporting

features. Microsoft Project allows for extensive customization of Gantt charts and provides various views to analyse project progress.

Asana: Asana is a popular cloud-based project management software that offers Gantt chart functionality through its Timeline feature. Users can create tasks, set dependencies, and visualize project timelines in a Gantt chart format. Asana's Gantt charts are interactive and allow for easy drag-and-drop scheduling of tasks. Teams can collaborate in real-time and track project progress directly within Asana's user-friendly interface.

Trello: Trello is a popular Kanban-style project management tool that offers Gantt chart capabilities through third-party integrations such as Placker and Ele Gantt. These integrations allow Trello users to visualize their Trello boards as Gantt charts, providing a high-level overview of project timelines and dependencies. Trello's card-based system makes it easy to organize tasks and manage project workflows, while the Gantt chart integration enhances project planning and coordination.

Smartsheet: Smartsheet is a collaborative work management platform that includes Gantt chart functionality as one of its core features. Users can create Gantt charts by adding tasks, durations, dependencies, and milestones to their Smartsheet projects. Smartsheet offers robust collaboration tools, such as comments and attachments, to facilitate communication among team members. The platform also supports automated alerts and reminders to keep projects on track.

Monday.com: Monday.com is a versatile project management tool that provides Gantt chart views to visualize project schedules and dependencies. Users can create tasks, assign owners, and set deadlines within Monday.com intuitive interface. The Gantt chart view allows teams to identify critical paths, monitor progress, and make adjustments to project timelines as needed. Monday.com also offers integration with various third-party apps to streamline workflow management.

TeamGantt: TeamGantt is a dedicated Gantt chart software that focuses on simplicity and ease of use. Users can create Gantt charts online, set dependencies, and collaborate with team members in real-time. TeamGantt is popular among small to mid-sized teams for its intuitive interface.

Wrike: Wrike is a cloud-based project management software that offers Gantt chart functionality for visualizing project timelines and dependencies. Users can create tasks, assign dependencies, and track progress directly within Wrike's platform. Wrike's Gantt charts support drag-and-drop functionality, allowing for easy adjustment of task durations and deadlines. The software also offers advanced reporting and analytics features to help teams gain insights into their project performance.

2.3 Gantt Chart Framework

Collaboration and Communication both are important at same time for successful implication of Gantt chart in project management. Gantt chart give the mutual understand as well as ensure that every stakeholder understand their key responsibilities towards the project Genua et al. (2006). Furthermore, every team member takes the accountability of their assigned tasks and communicate with team members timely if need any support. To see the project progress, project manager can do regular meetings and stakeholders can discuss any challenges faced in project. By incorporating these daily practices, organizations can make Gantt chart a powerful tool for project collaboration and communication which could led to successful project outcomes. There are two methods to creating a Gantt chart Özdamar and Ulusoy, (1996). They are:

- 1) **Forward Schedule:** Starts with the project start date and moves forward in time until you hit the project deadline.
- 2) **Backward Schedule:** Begins with the project deadline and works backward to determine when each activity should start in order to meet the deadline.

On behalf of my research question, I would say integrating Gantt charts into the project activities and communication tools will improve the coordination, collaboration and transparency among internal and external team members. There are detailed steps to create a Gantt chart, once the understanding of the steps is cleared, there is feature which include project scope, objectives, timelines, gathering information, prioritizing tasks, identify task durations, alien the tasks and prepare Gantt chart template. Genua et al. (2006) say that this collaboratively make the Gantt chart a communication tool. To create a Gantt chart, it's essential to understand its key steps Valentin Huang, (2023). Here are the critical steps in Table 1 below:

Table 1. Key elements to create Gantt chart

S. no	Steps	Description
1	Define Project Scope and Objectives	Before creating a Gantt chart, clearly define the project scope and objectives. This ensures that all necessary tasks are included in the chart.
2	Establish the Project Timeline	Determine the project's start date and end date to establish the overall time frame for the project.
3	Gather Activity Information	Compile all necessary information regarding the activities involved in the project. This may involve utilizing a Work Breakdown Structure (WBS) to break down the project into smaller, manageable tasks.
4	Identify and Prioritize Tasks	Break down the project into smaller tasks and identify their dependencies. Each task should be specific, measurable, achievable, relevant, and time- bound (SMART). Then, prioritize these tasks based on their importance and sequence.
5	Estimate Task Durations	Estimate the time required to complete each task. This helps in scheduling and resource allocation. Be realistic with your estimates to ensure accuracy in the Gantt chart.
6	Sequence Tasks	Determine the order in which tasks need to be completed based on their dependencies.

		Some tasks may be sequential, while others can be done concurrently. Ensure that tasks are logically sequenced to avoid delays.
7	Create the GANTT Chart Template	Construct the basic framework of the Gantt chart, including a timeline representing the project duration and a list of activities.
8	Schedule Activities	Utilize either Forward Scheduling or Backward Scheduling techniques to add bars to the Gantt chart representing each activity. Ensure that dependency are accurately depicted, and the full duration of each activity is accounted for.

2.4 Role of Gantt Chart in PM Methodology

Gantt charts have long been considered a fundamental tool for project management. A Gantt chart visually displays a project's timeline and the tasks that need to be completed within that timeline, making it easy for project managers to see at a glance the progress of a project. However, Gantt charts are only one tool in a project manager's toolbox. In order to effectively manage a project, project managers often use a variety of tools, including task management tools, resource management tools, and collaboration tools. Integrating these tools alongside a Gantt chart offers project managers a comprehensive perspective of their project, streamlining their workflow by eliminating the necessity to switch between various platforms Pankaja Pradeep Kumar (2005).

1. Waterfall Methodology

In the traditional waterfall approach, leader's approaches to challenges like tasks are completed sequentially, with each phase (such as requirements gathering, design, development, testing, and deployment) following the previous one. It's suggested by Haque et al. (2017) that Gantt charts are excellent tools for illustrating the linear progression of tasks over time. Each task is represented as a bar on the chart, showing its start and end dates. Dependencies between tasks can be indi-

cated, ensuring that tasks are completed in the correct order. For example, development cannot begin until the design phase is complete. Project managers can use Gantt charts to visualize the entire project timeline, allocate resources, and track progress.

2. Agile Methodology

Agile methodologies, like Scrum, emphasize iterative development, flexibility, and collaboration. While Thesing et al. (2021) say that Agile teams primarily use tools like Kanban boards or sprint backlogs for day-to-day task management, Gantt charts can still be useful for high-level project planning and visualization. Agile teams can create Gantt charts to plan iterative development cycles (sprints), identify dependencies between user stories or tasks, and communicate timelines to stakeholders. However, Gantt charts in Agile environments may need to be more flexible and subject to change compared to traditional project management approaches.

3. Critical Path Method (CPM)

CPM is a project management technique used to identify the longest sequence of dependent tasks (the critical path) and determine the shortest possible duration for completing a project. Gantt charts can be used to visualize the critical path. According to Yamín and Harmelink (2001) Critical tasks are those that, if delayed, would impact the overall project timeline. By highlighting critical tasks on the Gantt chart, project managers can focus their efforts on managing these tasks to ensure the project stays on schedule.

4. Lean Project Management

Lean project management focuses on maximizing value while minimizing waste Eric Gabriel (1997).

Gantt charts can be used in lean environments to:

- Visualize value stream mapping: Identify the flow of work, bottlenecks, and areas for improvement.
- Identify bottlenecks: Highlight tasks that slow down the overall process.
- Optimize workflows: By mapping out tasks and their dependencies, lean teams can identify opportunities for improving efficiency and reducing waste.

5. PRINCE2 (Projects IN Controlled Environments)

PRINCE2 is a structured project as said AXELOS (2017) and management methodology that divides projects into stages (initiation, planning, execution, monitoring, and closure).

Gantt charts can be used within PRINCE2 to:

- Plan and visualize project stages.
- Emphasize control over the project's lifecycle.
- Communicate project timelines to stakeholders.

2.5 Effective Communication with Gantt Chart

Gantt charts are the most familiar communication tool which offer features like a task assignment, tracking the progress, and notify the milestones. Effective communication shown by Gantt by assign tasks, regular updates on tasks, and receive the alters for upcoming deadlines. Also, for the remote work environment its best tool for communication with productive results and better accountability Z.A. Aziz, (2021). Effective communication and collaboration through Gantt chart can overcome the geographical barriers and can win projects. It's a powerful tool in terms of many industries using like construction, marketing, event planning, product development etc. If organizations use Gantt chart, it become the real-time medium and support, promote internal and external communication, enhance transparency and help remote collaboration Chiochio et al. (2012).

Gantt charts provide a platform for real-time communication, Nuño et al. (2018) say keep all the team members connected and engage for running project. One can communicate with each other through instant messages or can directly comment on the comment threads with the chart. Real time collaboration helps in quick decision making and keep all stakeholders aligned with the project timelines and goals. Also, challenges are the part of project management, where Gantt chart become powerful tool to handle project as visual representation with proper timelines, tasks, dependencies and showcase the clear picture to the entire team. This concept is called team EAM alignment where everyone remains informed and engaged about the project. To build accountability within the team everyone must focus on transparency. Culture of open communication within the team, one must need project related document within chart Koskinen (2024). In the digital world, the remote collaboration has become common after the COVID-19 pandemic. Gantt chart fully support the remote collaboration which give easy in providing a centralized platform for remote-communication Vasankari, Peetu, (2024). This centralized approach reduces the communication gaps, promote collaboration and ensure to keep team members connected and engaged till the completion of the project.

3 RESEARCH METHODOLOGY

This chapter describes the research methodology used in the study. Methodology means the research design, data collection methods, data analysis procedures, and the justification for choosing these methods. The thesis aims to ensure the reliability and validity of the results obtained from Interviewers. Qualitative research is particularly suited for experiencing a deep understanding of complex phenomena through rich, detailed data Creswell (2013).

This study focuses on understanding the lived experiences of individuals using Gantt charts in their daily life project management practices. According to Moustakas (1994), phenomenology is appropriate for exploring the essence of experiences from the perspectives of those who have directly encountered the phenomenon.

Qualitative Data Analysis (Data Collection) proceeds as follows: The interview transcripts were analysed using thematic analysis, a method for identifying, analysing, and reporting patterns within data (Braun & Clarke, 2006). The process involved analysing the data, generating themes, and reviewing these themes to ensure they accurately represented the researched questions. This method allowed for a detailed understanding of the participants' experiences and perspectives regarding the use of Gantt charts.

The primary data collection method used in this study was structured interviews. Total of four project managers with extensive experience using Gantt charts were interviewed from different regions and companies. This method allows for flexibility in exploring participants' experiences while ensuring that key topics are covered (Kvale & Brinkmann, 2009). Each interview lasted between 40 and 45 minutes and was conducted via video conferencing.

3.1 Evaluation of Reliability and Validity

For more credible results in case study, it's important to ensure reliability and validity. Reliability means to the consistency of the research process, while validity refers the accuracy of the findings. The following measures were taken:

Construct validity relates to accurately representing the concepts being studied. This was achieved through:

- Triangulation: In this thesis multiple sources of evidence, including interviews with three project management professionals (Interviewees A, B, and C) from different industries, to verify diverse perspectives.
- Chain of Evidence: Support a clear chain of evidence, such as corroborating Interviewee A's use of Agile methodology in healthcare projects with Interviewee B's use in ERP implementations (Yin, 2003).

Internal validity concerns the accuracy of causal connection identified. This was addressed by:

- Pattern Matching: Comparing empirical patterns from interviews with predicted patterns from literature. For example, the effective use of Gantt charts across interviews aligns with established project management practices Miles & Huberman (1994).
- Explanation Building: Emerge theoretical explanations based on interview data, such as Interviewee C's use of Primavera P6 in construction, linking specific tools to project success Yin (2003).

External validity refers to the generalizability of findings. This was ensured by:

- Replication Logic: Conducting interviews across different industries to ensure findings are not unique to a single context. Consistent use of methodologies and tools suggests broad applicability.

Reliability concerns the consistency of the research method. To ensure reliability can be ensured using the following techniques:

- Case Study Protocol: Using a standardized interview protocol for consistency in data collection.
- Case Study Database: Creating a database of transcripts and notes to allow for replication by other researchers.

By implementing these measures, the research ensures robust construct validity, internal validity, external validity, and reliability, thereby strengthening the credibility and generalizability of the findings.

3.2 Case Study

Case study approach was employed to take a comprehensive understanding of the use of Gantt charts in project management within real-world contexts. A case study itself is an in-depth, detailed examination of a subject (the case), as well as its related contextual conditions. It's particularly suited for examining contemporary phenomena in depth and within their real-life settings (Yin, 2018). It allows focusing on specific instances within different companies' sizes. This approach allows for a detailed exploration of how Gantt charts are utilized, the challenges encountered, and the perceived benefits from the perspectives of both project managers and CEOs.

Collecting all the data involved in-depth, defined structured interviews with each participant, allowing for the exploration of their experiences and perceptions regarding Gantt charts. The interviews were recorded, transcribed, and analysed using thematic analysis. This analysis focused on identifying key themes related to the effectiveness, challenges, and impact of Gantt charts on project management processes. By examining these themes within the context of each case, the study provides excellent insights into how Gantt charts are applied and perceived in different organizational settings. This case study approach not only enhances the

depth of understanding but also contributes to the validity of the research findings by situating them within real-world practices.

3.3 Empirical Findings

To find out case study insights, multiple key personnel from three companies, BenmarGroup (USA), Scania (Sweden) and AtkinsReális (London), were interviewed to gain insights from their multiple roles like CEO, Project Manager, and Project Control. The topics particularly focused on were Gantt Chart integration, managing timelines, task dependencies, collaboration, communication, and identifying limitations and advantages. I personally reached out to various companies from different regions and carefully selected three multinational companies to interview them. Interviewee A was conducted online with the CEO and the Project Manager of a large company in the USA; Interviewee B was conducted online with the Project Manager in Sweden; and Interviewee C was conducted online with the Project Controls Engineer of a large company in London.

Formal emails were sent to all companies to schedule the interviews, and the dates were confirmed promptly by the key personnel. I conducted all my interviews online in English and they were recorded to ensure the accuracy and reliability of the data collected. The interviews were started with an introduction, followed by the structured approach and communicating the research questions. However, all the interviews were up to 45 minutes in duration. Appendix 1 reflects all the structured questions prepared for the interviewers and insightful from these interviewers are mentioned in detail. The details of the interviews are shown in Table 2.

Table 2. Details of interviewed key personnel

Company	Region	Work Position	Year of Experience	Interview Dates
Interview A: BENMARGROUP	USA	Interview A1: CEO	25 Years	13 June, 2024
		Interview A2: Project Manager	6 Years	13 June, 2014
Interview B: SCANIA	Sweden	Project Manager	10 Years	14 June, 2024
Interview C: ATINSREALIS	London	Project Controls Engineer	11 Years	14 June, 2024

3.4 Interviews and Data Collection

This chapter presents the findings from the interviews conducted. All the interviews were conducted in the same sequence covering all the important topics. The theoretical framework started with general project management approach, Gantt chart usage, Gantt chart in project collaboration, and Gantt chart in communication.

3.4.1 Interview A

With Interviewee A, the discussion proceeded on overall approach to project management. According to him, various frameworks such as Agile and Waterfall methodologies are used in project management which depends on specific project needs *“Our process project management process involves various frameworks and project management methodology is one of the frameworks. For large implementation projects, we apply a formal project management structure that details*

phases, activities, and sub-tasks. This encompasses timelines, people, processes, and technology.”

Interviewee A talked about project management methodologies that they typically use in their company. He says, because of the fact that most of implementations are long, and involves a lot of people, there are times they have 200 - 250 people: *“For well-defined projects like implementing Oracle Applications in HR, we use the waterfall method due to its sequential nature. For product development with undefined tasks, we use agile, building and testing each feature iteratively. Most of our projects are large, well-defined, and involve many people, making waterfall the preferred approach. “Example” in a healthcare project, a hospital needed a custom barcode system for their inventory. We used agile because the requirements were unique and evolving. We defined each feature, created the code, tested it, and validated it with the customer before moving to the next feature”.*

Interviewee A mentioned about the tools or software they use for project planning and scheduling. He says majority of the client also have their own preferences but they preference is for Microsoft Project as a tool: *“Most of our large clients, like the University of California and many hospital systems, prefer Microsoft Project for both agile and waterfall methods. Smaller clients might use Asana, SharePoint, or ServiceNow. Clients prefer sticking to one tool to simplify training for their team members.”*

Interviewee A emphasises the planning and scheduling for a new project. He says *“We just started a project with the University of California. First, we validate the stakeholders' objectives, develop a scope document, and create a project charter. We identify both our team and the client's team for collaboration. In the first 4-8 weeks, we develop comprehensive documentation, including a project plan, communication plan, risk plan, issues plan, testing plan, go-live plan, and change management plan. We establish communication protocols and define roles from the*

project team to the executive sponsor. This serves as the project's architecture for the next 1-2 years”.

The interview was extended to include a discussion on the primary benefits of using Gantt charts for project management. According to Interviewee A, the Gantt Chart is incredibly and beneficial for several reasons. *“First, Gantt chart provide a clear visualization of the entire project schedule, making it easy for everyone to understand timelines, task dependencies, and deadlines. Second, Gantt chart enhance task management by breaking down the project into smaller tasks, assigning them to team members, and improving overall clarity. Lastly, Gantt charts improve progress tracking by allowing us to quickly identify and address any difficulties or errors. These benefits make Gantt charts essential for effective project management”.*

To the question how effective is the Gantt chart for complex project timelines, the CEO answered: *“We find Gantt charts extremely effective for summarizing complex project timelines. At UPMC, we had a massive project involving 24 hospitals to eliminate paperwork in procurement. The detailed project plan spanned over 100 pages, which was overwhelming for our executives and stakeholders. Using a Gantt chart, we provided a high-level overview of the process, plan, timeline, and major dependencies, cutting through confusion. It allowed everyone to quickly grasp key aspects without getting lost in the details, making it an essential tool for clear communication and effective project management”.*

Interviewee A said that there is one example where the Gantt chart significantly improved the team collaboration: *“We had a project with the University of California system involving eight external vendors providing software and applications. By using a Gantt chart, we simplified the presentation of the overall timeline, dependencies, and critical paths. This approach enhanced collaboration by clearly communicating project milestones without sharing extensive internal details, facilitating smoother coordination with external vendors”.*

When discussing the remote collaboration, he said: *“Gantt charts and project management plans are crucial in remote work environments. Previously, we could physically check progress in team offices or on large boards. With most people now working from home, it's essential to rely on unified tools and plans. Formality in processes has become more critical, ensuring everyone has access to the same project plan and Gantt chart. This consistency is vital for visualizing impact and maintaining effective communication in remote settings, which has become more complex since the shift to remote work”*.

When asked about the integration of Gantt chart into the project management process, Interviewee A said: *“We use Microsoft Project because it offers features for creating Gantt charts, but it's not the simplest tool due to its complexity over the years. Clients often export essential data to create more user-friendly Gantt charts using simpler, visually enhanced tools online, which offer color coding and 3D features for better visualization”*.

Talking about a Gantt chart feature, Interviewee A says *“the simplicity and uniformity of Gantt charts most useful, emphasizing their practicality and cautioning against excessive customization that may reduce usability and undermine their seriousness”*.

Interviewee A has encountered also limitations: *“Gantt charts where some individuals prefer detailed breakdowns rather than high-level overviews, which can obscure the reasons behind timeline changes or dependencies”*.

When describing communication as a term Interviewee A Project Manager *“We see communication as a two-way process where information and ideas are exchanged to foster better understanding. Effective communication within a team is pivotal for success, particularly in remote work scenarios where timely responses are crucial. It serves as the cornerstone of successful project management, ensuring everyone is aligned and working towards common goals”*.

3.4.2 Analysis of Interview A

Interviewee A recognised the role of Gantt chart in project communication and collaboration. Table 3 presents a summary analysis of the interview.

Table 3. Analysis of interview A

Analysis of Interview A	
Project Management Approach and Tools	Interviewee A provides insights into their organization's robust approach to project management, leveraging various methodologies tailored to project specifics. For large-scale implementations like Oracle Applications in HR, the waterfall method is favoured due to its sequential nature, ensuring structured progress through defined phases. Conversely, agile methodologies are employed for projects with evolving requirements, such as custom barcode systems in healthcare, allowing iterative development and customer validation.
Tools and Software for Project Planning	The preference for Microsoft Project among larger clients like the University of California underscores the need for comprehensive project management tools capable of accommodating both agile and waterfall methodologies. This choice streamlines project planning and scheduling, promoting consistency and facilitating team training across diverse project scopes and complexities.
Planning and Scheduling Approach	Interviewee A emphasizes meticulous planning at project initiation, focusing on stakeholder alignment, scope development, and detailed documentation. This

	structured approach, spanning initial documentation phases to communication and risk management plans, lays a robust foundation for long-term project success and collaboration between internal and client teams.
Benefits of Gantt Charts	Gantt charts are highlighted as pivotal tools for visualizing project schedules, dependencies, and milestones, enhancing task management and progress tracking. Interviewee A emphasizes their role in simplifying complex timelines, such as the extensive project involving 24 hospitals at UPMC, by providing executives and stakeholders with clear, high-level overviews crucial for informed decision-making and effective communication.
Impact on Team Collaboration and Remote Work	In remote work environments, Gantt charts and structured project management plans serve as essential tools for maintaining visibility, ensuring consistent communication, and mitigating the challenges of dispersed teams. Interviewee A underscores the importance of unified tools and formalized processes in facilitating seamless remote collaboration, aligning team efforts, and sustaining project momentum across virtual settings.
Integration and Limitations	While Microsoft Project integrates Gantt chart functionalities, Interviewee A acknowledges complexities that may necessitate simpler, more visually enhanced tools for broader team accessibility and comprehen-

	sion. Challenges include balancing detailed breakdowns with high-level overviews to preserve transparency and clarity in project timelines and dependencies.
Communication and Project Success	Effective communication emerges as a cornerstone of successful project management in Interviewee A approach. Clear, two-way communication fosters understanding, alignment, and timely responses among team members, crucial for navigating complexities in both traditional and remote work environments.

3.4.3 Interview B

When Interviewee B was asked about their overall approach to Project Management in their organization, she said: *"We blend the PPS model with a mix of agile and waterfall methods. We start by aligning processes in kick-off meetings and then dive into implementing D365 and integrating additional features from independent software vendors. During the discovery phase, we identify gaps in ERP implementation, such as specific country regulations. Finally, we prepare the project by outlining scope, estimating efforts, and gaining approval from regional and local CFOs"*.

Interviewee B talks about project management methodology they use mostly: *"For ERP systems, we combine waterfall with agile methodologies. We begin with waterfall for discovery and feasibility phases, ensuring thorough planning and alignment. During execution, we adopt agile, specifically using sprint cycles in a Scrum framework. This approach allows us to iterate monthly, ensuring we meet timelines and adapt as needed based on project progress"*.

Concerning project planning and scheduling, Interviewee B were asked to tell what tools and software they use: *"Previously, we used JIRA, but it didn't align well with*

our Microsoft ERP and Azure deployment pipeline. As a result, we migrated our projects to Azure DevOps, where we now manage the entire project lifecycle. Azure DevOps provides us with comprehensive insights into deployment pipelines and allows us to track changes effectively".

About planning and scheduling new project Interviewee B says: "In Azure DevOps, we recently integrated the Portfolio Plus add-on, which functions as a robust tool for timelines and Gantt charts, seamlessly complementing our project management needs within Azure. While JIRA also offers similar add-ons, our decision to migrate to Azure led us to select Portfolio Plus after extensive research. This tool enables us to create comprehensive timelines from the project's initial phase through execution, user testing, and hyper care stages. Aligned with our PPS methodology, our project planning includes Decision Point meetings held monthly, featuring local steering led by the country's CFO. These meetings review achievements, identify risks, and make crucial decisions across seven milestone Decision Points throughout the project lifecycle".

Interviewee B said that the primary benefit of using the Gantt chart: "Gantt charts are crucial for visualizing project timelines effectively. We found them indispensable in Azure DevOps, leading us to adopt the Portfolio Plus add-on. They provide a clear, visual representation that simplifies understanding of project progress, dependencies, and upcoming tasks, enhancing overall project management clarity and communication".

On the case specific questions on complex project timelines Interviewee B says: "Gantt charts are invaluable for managing our current complex project, involving dependencies across multiple teams and external integrations. They allow me to clearly communicate timelines for testing, configurations, and other key milestones to stakeholders and team members. This clarity helps everyone involved plan their tasks and capacities accordingly, ensuring we meet deadlines despite potential deviations, especially in projects spanning up to three years, varying by country due to regulatory complexities".

Interviewee B did not mention a project but showed her point of view on team collaboration: *"Gantt charts have significantly improved our team collaboration by clearly assigning tasks, deadlines, and workload visibility. For instance, they enable us to coordinate effectively, ensuring everyone understands their roles and responsibilities, which fosters cohesive teamwork and timely project delivery"*.

Interviewee B says that Gantt charts supported remote collaboration, especially with the increase in remote work: *"She emphasized that Gantt charts are pivotal for remote team collaboration, particularly given their globally dispersed team members. Regardless of COVID-19, her team operates across different continents, making visualizations essential for aligning on project goals and individual responsibilities. Gantt charts enable clear understanding of task assignments, dependencies, and deadlines, facilitating effective communication and ensuring timely deliverables across diverse time zones"*.

Regarding integrating Gantt chart Interviewee B said: *"We've integrated Gantt charts into our project tool and team's channel add-ons. These charts are accessible to stakeholders on Microsoft Teams, allowing real-time updates and ensuring everyone stays informed about timeline changes and project progress consistently"*.

Interviewee B highlighted specific feature of Gantt chart: *"She say we find the effort estimation feature particularly valuable, using story points to estimate feature development times. This helps us communicate clear expectations to end users, ensuring they allocate adequate time for testing and reducing surprises during project demos"*.

Interviewee B continued with limitations: *"Personally, she finds the main limitation is the manual effort required when there are deviations or changes in plans. Ensuring that all related tasks and dependencies reflect these changes can be time-consuming and prone to errors, especially adjusting dates across multiple tasks manually"*.

Interviewee B talked about the communication term: *"She say in our projects, communication is crucial due to numerous external dependencies. Visualized charts like Gantt charts help illustrate dependencies clearly, ensuring tasks are remembered and commitments are visible in everyone's backlog, aligning communication with the actual project plan"*.

When discussing the communication affecting project performance in detail, Interviewee B: *"before using Gantt charts and structured tools, ensuring external dependencies were met on time was difficult. Agreements made in meetings sometimes got overlooked, leading to delays. Now, with Gantt charts and clear deadlines, it's easier to track commitments and ensure tasks are completed on schedule, significantly improving project management efficiency compared to before"*.

Interviewee B continues: *"Gantt charts enhance project performance by providing a clear visual timeline that aids in tracking progress, managing deadlines, and identifying dependencies. This organized approach supports effective planning and resource management, crucial for achieving timely project completion and success"*.

Interviewee B point of view on Gantt: *"Using Gantt charts significantly improves communication efficiency by reducing the time spent in meetings and streamlining updates through tools like Azure DevOps. Notifications ensure everyone is informed promptly, facilitating clearer and more direct communication about changes, tasks, and deadlines without the need for frequent meetings"*.

3.4.4 Analysis of Interview B

Interviewee B acknowledged how the Gantt charts facilitating the scheduling approach and project planning. Table 4 provides a summary analysis of the interviewee B.

Table 4. Analysis of interview B

Analysis of Interview B	
Project Management Approach and Tools	Interviewee B organization uses a blend of the PPS model with agile and waterfall methods. Their approach starts with aligning processes in kick-off meetings, implementing D365, integrating additional features from independent software vendors, and identifying gaps during the discovery phase, such as country-specific regulations. The project preparation phase includes outlining scope, estimating efforts, and obtaining approval from regional and local CFOs.
Tools and Software for Project Planning	The organization previously used JIRA but switched to Azure DevOps to better align with their Microsoft ERP and Azure deployment pipeline. Azure DevOps now manages the entire project lifecycle, providing comprehensive insights into deployment pipelines and effective tracking of changes.
Planning and Scheduling Approach	In Azure DevOps, the organization integrated the Portfolio Plus add-on, which is robust for timelines and Gantt charts. This tool complements their project management needs, creating comprehensive timelines from the project's initial phase through execution, user testing, and hyper care stages. Monthly Decision Point meetings, led by the country's CFO, review achievements, identify risks, and make crucial decisions across seven milestone Decision Points throughout the project lifecycle.

Benefits of Gantt Charts	Gantt charts are crucial for visualizing project timelines effectively. Gantt charts offer straightforward visual that simplifies understanding project progress, dependencies, and upcoming tasks, thereby improving overall clarity and communication in project management.
Impact on Team Collaboration and Remote Work	Gantt charts have significantly improved team collaboration by clearly assigning tasks, deadlines, and workload visibility. They enable effective coordination, ensuring everyone understands their roles and responsibilities, fostering cohesive teamwork and timely project delivery. In remote work scenarios, especially with globally dispersed teams, Gantt charts are pivotal for aligning project goals and individual responsibilities, facilitating effective communication, and ensuring timely deliverables across diverse time zones.
Integration and Limitations	Gantt charts are integrated into project tools and team channel add-ons, accessible to stakeholders on Microsoft Teams for real-time updates. However, a significant limitation is the manual effort required when plans change. Adjusting dates across multiple tasks manually can be time-consuming and prone to errors.
Communication and Project Success	Communication is crucial due to numerous external dependencies. Visualized charts like Gantt charts help illustrate dependencies clearly, ensuring tasks are remembered and commitments are visible in everyone's backlog, aligning communication with the actual project plan. Before using Gantt charts, ensuring external

	dependencies were met on time was challenging. Now, with Gantt charts and clear deadlines, tracking commitments and ensuring task completion is easier, significantly improving project management efficiency. Gantt charts reduce meeting time and streamline updates through tools like Azure DevOps, enhancing communication efficiency by promptly informing everyone about changes, tasks, and deadlines without frequent meetings.
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3.4.5 Interview C

Interviewee C highlighted their company's overall approach to project management is integral to our operations: *"As a Project Management employee, I ensure that project management is deeply embedded within our core practices. Additionally, we emphasize the role of project controls, which include tasks like costing, budgeting, and scheduling, often managed by specialists who support project managers. Our approach involves using innovative methodologies, the right tools, and clear concepts to define and manage projects effectively, aiming to achieve the best outcomes"*.

Interviewee C mentioned about the Project Management Methodologies: *"In our construction engineering consulting company, we primarily use the Waterfall method, the Critical Path Method (CPM), and Earned Value Management (EVM). Agile and Scrum are more commonly used in the IT sector. Our projects, such as construction, have well-defined tasks from start to finish, making the Waterfall method suitable. CPM helps us schedule and track critical activities to ensure timely completion, while EVM allows continuous monitoring of costs, time, and budgets against the baseline"*.

Interviewee C talked about the tools or software used in the project planning and scheduling: *"We use three main tools for project scheduling: Excel, Microsoft Project, and Primavera P6. Excel is used for small projects to create basic schedules. For more complex projects requiring detailed breakdowns, we use Microsoft Project and Primavera P6. Primavera P6, an Oracle software, is particularly beneficial for large, complex projects due to its robust features, including both desktop and server versions, allowing multiple users to update schedules simultaneously"*.

Interviewee C emphasised on planning and scheduling new projects: *"We start with a kick-off meeting involving the project management team, project managers, and discipline leads. We identify and sequence all necessary activities, create a draft schedule using tools like Excel, Microsoft Project, or Primavera P6, and establish start and finish dates. After refining the schedule in follow-up meetings, we set a baseline and monitor progress through regular meetings, comparing current status against the baseline"*.

The Interviewee C described: *"The benefits of using Gantt charts are that you get a streamlined one-box view where you can see all the activities, the people responsible, and the timeline. With a glance, you can easily understand what activities need to be done in the next two or three weeks. The different coloured bars and the timeline make it simple to visualize the project. This helps project managers and schedulers to plan and monitor progress more effectively"*.

As a Project Controller, Interviewee C explained how Gantt charts in conveying complex projects timelines are managed: *"There are multiple tools integrated with commonly used Gantt charts, like MSP or Primavera P6, which offer Gantt chart visuals. These tools also provide other views where you can see the resources deployed for each activity, helping stakeholders understand the resource needs. Resources can be people, machines, or materials. These views allow us to see the type of resource required, its cost, and whether it's a fixed or variable cost, like the cost of using a crane on a construction site"*.

Interviewee C mentioned a project example and said: *"We recently completed a website redesign project for a client. The project included various phases: planning, design, development, testing, and deployment. By using Gantt charts, we could clearly outline each phase and identify task dependencies. This visual representation helped team members understand their roles and deadlines better, enhancing coordination and communication. Consequently, we managed to deliver the project ahead of schedule with fewer revisions, demonstrating how Gantt charts fostered improved collaboration among our team"*.

During the shift to remote work during COVID-19 pandemic, Gantt charts have become crucial, as Interviewee: *"Gantt charts have supported remote collaboration, especially with the increase in remote work, by allowing team members to update and access project timelines online. Before COVID-19, Gantt charts were often printed and discussed in person. Now, with tools like Microsoft Teams and Zoom, teams can share screens and discuss Gantt charts in real-time, making collaboration more efficient despite being remote. While Gantt charts themselves haven't changed, the way we use them in a remote environment has become more prevalent and streamlined due to the shift to online workspaces"*.

Interviewee C commented on Gantt chart integration: *"We integrate Gantt charts into our project management process by using them as a basic timeline view. However, for different stakeholders who need various perspectives, we often create custom views. For instance, some stakeholders prefer seeing critical paths highlighted in different colors and detailed steps within longer activities. To achieve this flexibility, we download the list of activities from the tool into Excel and then integrate it with other software like Microsoft Visio. Visio allows us to create complex network diagrams with multiple boxes and arrows, providing a detailed and customized view of the project"*.

Interviewee C emphasized specific features of Gantt charts, stating, *"The most useful aspects are the baseline and progress columns. The baseline allows me to compare the current status with the original plan, showing any shifts in activities."*

The progress column helps track how much work has been completed. Even if the project timeline has shifted, seeing a high percentage of progress can indicate that we are on track to meet deadlines”.

Interviewee C about limitations: *“Yes, Primavera P6, though excellent for complex projects, is not very user-friendly compared to Microsoft Office tools, requiring a significant learning curve. However, it remains well-suited for managing larger project sets”.*

Communication as a term described by Interviewee C: *“Communication, to me, is about clearly expressing project needs, progress, and plans. It involves having honest conversations about the project's status, addressing issues transparently, and effectively managing tasks to keep the project on track”.*

Interviewee C commented on: *“Effective communication, facilitated through regular scheduled meetings with planners and project managers, is crucial for project performance. Without it, working in silos can lead to misunderstandings and inefficiencies, impacting overall project outcomes significantly”.*

Cross questioned and Interviewee C answered: *“The Gantt chart enhances communication quality by providing a visual tool that clarifies project timelines and dependencies, ensuring better coordination among team members and stakeholders”.*

3.4.6 Analysis of Interview C

Interviewee C highlighted the significance of Gantt charts for integration of chart and limitations for project size with Gantt chart. A summary of the interview analysis is presented in Table 5.

Table 5. Analysis of interview C

Analysis of Interview C	
Project Management Approach and Tools	Interviewee C emphasized their company's dedication to integrating project management deeply within their operations, employing methodologies like Waterfall, Critical Path Method (CPM), and Earned Value Management (EVM) tailored for construction projects. They prioritize project controls such as costing, budgeting, and scheduling to ensure effective project management and optimal outcomes.
Tools and Software for Project Planning	They utilize a range of tools for project scheduling, including Excel for basic schedules, Microsoft Project for detailed breakdowns, and Primavera P6 for managing large, complex projects. Primavera P6 is highlighted for its robust features and ability to support multiple users simultaneously across desktop and server versions.
Planning and Scheduling Approach	Interviewee C begins projects with comprehensive kick-off meetings involving project management teams and discipline leads. They sequence activities, draft schedules using Excel, Microsoft Project, or Primavera P6, establish baselines, and monitor progress through regular meetings to ensure alignment with project goals.
Benefits of Gantt Charts	Gantt charts are praised for providing a clear visual representation of project activities, responsibilities,

	and timelines. They enable quick understanding of upcoming tasks and progress, enhancing project managers' ability to plan and monitor projects effectively.
Impact on Team Collaboration and Remote Work	Amid the shift to remote work during COVID-19, Gantt charts have supported collaboration by enabling online access and updates. Teams now utilize platforms like Microsoft Teams and Zoom to discuss Gantt charts in real-time, improving efficiency in remote project management.
Integration and Limitations	Interviewee C integrates Gantt charts into their project management process as fundamental timeline views. They create custom views to meet various stakeholders' needs, such as highlighting critical paths and tracking progress. However, tools like Primavera P6 are noted for their complexity compared to more user-friendly options, posing a learning curve for new users.
Communication and Project Success	Communication is described as essential, facilitated through structured meetings to ensure transparency and alignment across project teams. Effective communication prevents silos and misunderstandings, crucial for maintaining project efficiency and achieving successful outcomes.

3.4.7 Summary of Interviews

Based on the insights gathered from Interviews A, B, and C, each interviewee provided valuable perspectives on project management methodologies, tools, and

their approach to handling project complexities. Here's a synthesized conclusion in Table 6 based on their responses.

Table 6. Interviews summarization

Aspect	Interview A	Interview B	Interview C
Approach	Agile/Waterfall blend	PPS model with Agile	Waterfall, CPM, EVM
Methodologies	Agile/Waterfall	Agile/Scrum, Waterfall	Waterfall, CPM, EVM
Tools and Software	Microsoft Project, Excel for Planning	Azure DevOps, Portfolio Plus	Excel, Project, Primavera P6
Planning and Scheduling	Comprehensive, Structured	Azure DevOps, Milestones	Excel/Project/P6, Baseline
Gantt Chart Benefits	Visual, task clarity	Timelines, dependencies	Streamlined visualization
Complex Projects	Simplified timelines	Managing dependencies	Enhanced coordination
Remote Collaboration	Crucial for alignment	Essential for global teams	Supported through tools
Integration	Custom views, Visio	Microsoft Teams, Azure	Basic timeline, Visio

Useful Features	Baseline and progress tracking	Effort estimation with story points	Baseline and progress tracking for monitoring
Limitations	Detailed breakdowns	Manual effort	User-friendliness
Communication Approach	Two-way, critical Path	External dependencies	Transparent communication, scheduled meetings
Project Performance	Efficient task management	Timely task completion, resource management	Improved coordination, project efficiency

We have already discussed the approach and methodology in (section 2.4). Interviewee A emphasized a hybrid approach blending Agile and Waterfall methodologies, allowing flexibility depending on project requirements. They focus on comprehensive planning, stakeholder alignment, and the use of specific tools to ensure project success. However, Interviewee B described a blend of the PPS model with Agile and Waterfall methods, particularly for ERP projects, focusing on alignment during kick-off meetings and detailed planning during discovery phases. Interviewee C highlighted a structured approach primarily using Waterfall methodologies like CPM and EVM, particularly suited for well-defined projects in construction and engineering consulting.

Tools aspects were discussed in (section 2.2) Interviewee A prefers Microsoft Project for both Agile and Waterfall methods, with smaller clients using tools like Asana, SharePoint, or ServiceNow. Interviewee B moved from JIRA to Azure DevOps, finding it better aligned with their Microsoft ERP and Azure deployment

pipeline. Interviewee C uses Excel, Microsoft Project, and Primavera P6 for scheduling, with Primavera P6 preferred for its robust features for large, complex projects.

Interviewee A starts with validating stakeholder objectives, developing scope documents, and creating comprehensive project plans in the first 4-8 weeks. Interviewee B integrates the Portfolio Plus add-on in Azure DevOps for timelines and Gantt charts, holding monthly Decision Point meetings for review and decision-making. Interviewee C begins with kick-off meetings to identify and sequence activities, creating draft schedules and setting baselines for monitoring progress.

According to Interviewee A, the Gantt chart beneficial for visualizing project schedules, managing tasks, and tracking progress. Interviewee B emphasizes their utility in providing clear visual representations of project timelines, simplifying understanding of progress and dependencies it showed with example in (section 3.1). Interviewee C highlights their use in streamlining project timelines, managing resources, and enhancing communication among stakeholders.

Interviewee A noted a project with the University of California system where Gantt charts enhanced collaboration with eight external vendors by simplifying the presentation of timelines and dependencies. Interviewee B emphasized improved coordination and role clarity within their team through Gantt charts. Interviewee C shared an example of a website redesign project where Gantt charts helped deliver the project ahead of schedule with fewer revisions due to better team coordination. Interviewee A integrates Gantt charts via Microsoft Project, sometimes exporting data to more user-friendly tools. Interviewee B integrates them within Azure DevOps, accessible to stakeholders via Microsoft Teams. Interviewee C uses them as a basic timeline view, creating custom views for different stakeholders when needed.

All three interviews underscored that effective communication is vital for project performance as mentioned in section 3.2. Interviewee A finds Gantt charts essential for clear communication and effective management. Interviewee B noted that structured tools like Gantt charts have significantly improved efficiency and reduced delays. Interviewee C emphasized that visual tools like Gantt charts clarify timelines and dependencies, enhancing coordination and reducing misunderstandings.

4 CONCLUSIONS

4.1 Theoretical Contribution

RQ: How can Gantt charts be integrated in project collaboration and communication tools?

Gantt charts have stood the test of time, remaining a favoured tool in management despite their origins dating back over a century. Initially used for general production planning and control, they have evolved to serve project management primarily. This shift occurred because earlier attempts to apply Gantt charts to broader production issues faced significant challenges and were surpassed by advancements in technology. Today, with the advent of powerful computing tools, more sophisticated techniques are available for modelling complex production problems.

Gantt charts persist in project management because they offer a straightforward and accessible way to visualize crucial information. They serve as a user-friendly interface, allowing individuals to define problems clearly and comprehend proposed solutions more effectively. So, while newer technologies provide more advanced capabilities, Gantt charts remain valuable for their simplicity and practicality in project management contexts.

Here the study contributes to the theoretical understanding of project management tools by exploring the integration of Gantt charts within project collaboration and communication frameworks. After interviewing the project managers and CEOs from three companies, the findings provide valuable insights into the role Gantt charts play in enhancing project visibility, coordination, and communication. On the basis of theoretical understanding, the study extends existing theories on project management by highlighting how Gantt charts can be effectively incorporated into collaborative tools to streamline project processes and improve team

communication.

The results of the study suggest that Gantt charts, when integrated with any tools, can significantly enhance real-time project tracking and information sharing. This supports and extends the theoretical work of researchers like Kerzner (2017), who emphasized the importance of integrating project management tools with communication systems to facilitate better decision-making and project outcomes.

4.2 Practical Contribution

The study offers actionable insights for project managers and organizations looking to optimize their use of Gantt charts within collaborative and communication tools. The interviews highlighted several daily life best practices and strategies that can be implemented to enhance the effectiveness of Gantt charts in project management. For instance, integrating Gantt charts with project collaboration tools enables seamless updating and sharing of project timelines. Interviewer highlighted the importance of team collaboration.

These practical findings can serve as a guideline for other organizations aiming to achieve similar enhancements in their project management processes. The research underscores the importance of selecting the right integration tools, providing training to team members, and continuously monitoring and refining the integration process to address any emerging challenges.

4.3 Limitations and Future Research

While this study provides valuable contributions, it's important to acknowledge its limitations. The research is based on qualitative data from interviews with project managers and CEOs of three companies, which may limit the generalizability of the findings. Additionally, the study focused primarily on the integration of the Gantt charts with specific collaborative tools; future studies could explore the integration with other emerging project management technologies and platforms.

Another limitation is the reliance on self-reported data, which may be subject to biases such as social desirability or recall bias. Future research could incorporate quantitative methods, such as surveys or experiments, to validate and complement the qualitative findings. Longitudinal studies could also be conducted to examine the long-term effects of integrating Gantt charts with collaboration tools on project outcomes. Moreover, exploring the impact of such integration on different aspects of project management, such as risk management, resource allocation, and stakeholder engagement, could provide a more comprehensive understanding of the benefits and challenges associated with this approach.

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Appendix 1

Interview Questions in English (Less than 45 mins)

General Project Management

1. Can you describe your overall approach to project management within your company?
2. What project management methodologies do you typically use (e.g., Agile, Waterfall, Scrum)?
3. What tools or software do you use for project planning and scheduling?
4. How do you go about planning and scheduling a new project?

Gantt Chart Usage

5. In your experience, what are the primary benefits of using Gantt charts for project management?
6. How effective do you find Gantt charts in conveying complex project timelines to team members and stakeholders?

Gantt Chart in project Collaboration

7. Can you provide an example of a project where Gantt charts significantly improved team collaboration?
8. How have Gantt charts supported remote collaboration, especially with the increase in remote work?
 - How do you integrate Gantt charts into your project management process?
 - What specific features of Gantt charts do you find most useful?
 - Have you encountered any limitations or drawbacks in using Gantt charts?

Gantt Chart in Communication

9. How would you describe communication as a term?
10. How does communication affect project performance?
 - How does the use Gantt Chart affect project performance?
 - How does the use of Gantt chart affect the quality of communication?